Qwen3.8-27B NVFP4 + DSpark

RTX PRO 6000 Blackwell - Full benchmark, capacity, validation, and selected default configuration

119.19	7.68 ms	8 / 8	265.67
FULL-RUN OUTPUT TOK/S	MEAN TPOT	SUCCESSFUL REQUESTS	MAX OBSERVED DECODE WINDOW

Executive result

The exact published SGLang Qwen3.8 NVFP4 + DSpark stack was validated on one NVIDIA RTX PRO 6000 Blackwell Workstation Edition. The clean cookbook-shaped random workload sustained **119.188 output tokens/second**. A complete request reached 203.26 tok/s, and server decode windows exceeded the social post's 206.1 tok/s when DSpark acceptance was favorable. The selected configuration passed text, reasoning, native vision, structured tool-call, continuation, cold-start, and zero-restart gates.

Decision	Value
Selected default	RadixArk NVFP4 target + external DSpark draft under SGLang
Benchmark protocol	Random, ISL 8192, OSL 1024, 8 prompts, concurrency 1, cache flush
Isolation	Loopback-only measurement on 127.0.0.1:8010; no fleet traffic
Important boundary	Workload-sized memory ratio provides 196,618 KV tokens, not a full 262,144-token live pool

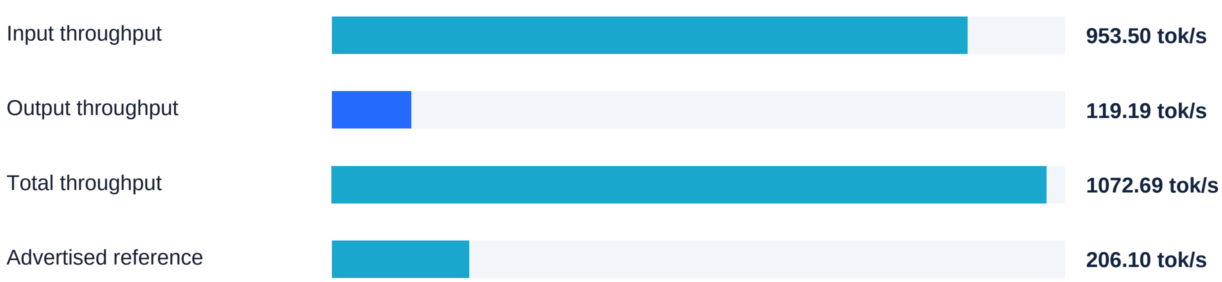
What DSpark is

DSpark is SGLang's external speculative draft model. It proposes multiple likely tokens and lets the 27B target verify them together. Performance therefore rises and falls with acceptance: repeated or predictable text can exceed 200 tok/s, while random continuations accept fewer draft tokens and run more slowly. DSpark is not the NVIDIA DGX Spark computer, and it is distinct from llama.cpp's built-in MTP implementation.

1. Benchmark workload and aggregate performance

Workload field	Exact value	Workload field	Exact value
Dataset	random	Backend	sglang-oai
Requests	8	Request rate	inf
Input/request	8,192 tokens	Output/request	1,024 tokens
Total input	65,536 tokens	Total output	8,192 tokens
Retokenized output	7,784 tokens	Range ratio	1.0
Configured concurrency	1	Effective concurrency	0.9997
Benchmark duration	68.7317 s	Request throughput	0.1164 req/s

Throughput



Input and total throughput share the same scale and are dominated by prefill. The 206.1 reference is a decode claim, so the directly comparable full-run result is the 119.188 output tok/s bar.

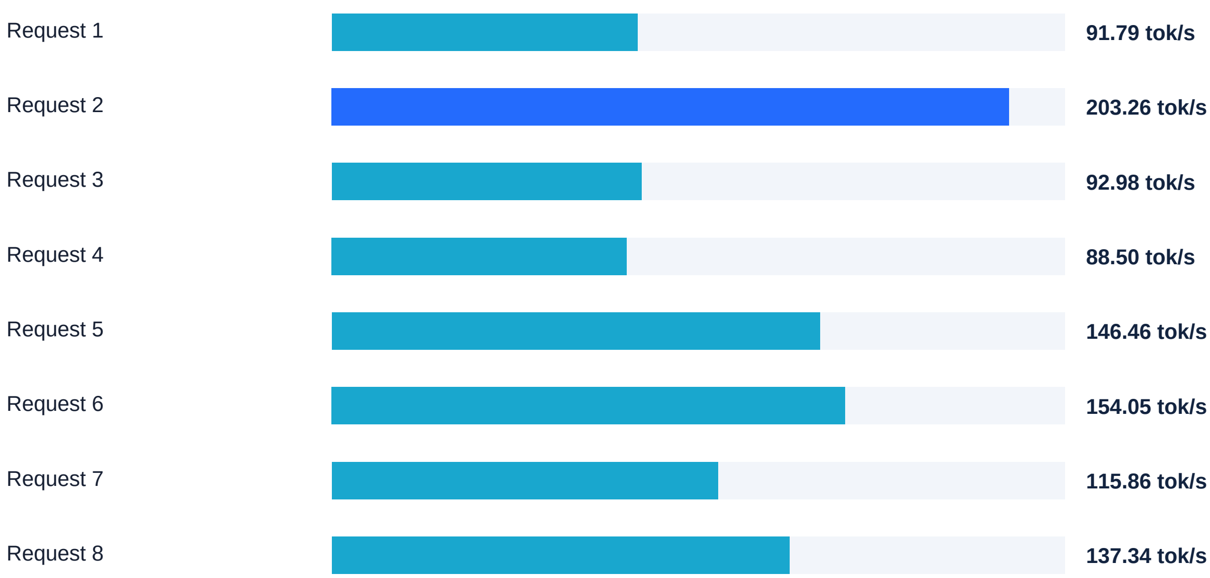
Latency distribution

Statistic	Mean	Median	Std dev	P90	P95	P99
E2E latency (ms)	8,588.84	8,147.20	2,281.37	11,280.61	11,425.50	11,541.41
TTFT (ms)	729.82	733.08	8.51	737.50	738.46	739.22
TPOT (ms)	7.682	7.243	2.233	10.325	10.460	10.568
ITL (ms)	7.949	4.840	7.085	19.157	19.265	38.274

Derived decode indicator	Rate
Mean TPOT equivalent	130.17 tok/s
Median TPOT equivalent	138.07 tok/s
Mean ITL equivalent	125.80 tok/s
Maximum observed ITL	57.43 ms

2. Per-request performance and acceptance sensitivity

Request	Input	Output	TTFT	E2E	Output rate
1	8,192	1,024	712.69 ms	11.156 s	91.79 tok/s
2	8,192	1,024	721.88 ms	5.038 s	203.26 tok/s
3	8,192	1,024	725.74 ms	11.013 s	92.98 tok/s
4	8,192	1,024	731.20 ms	11.570 s	88.50 tok/s
5	8,192	1,024	734.96 ms	6.991 s	146.46 tok/s
6	8,192	1,024	736.68 ms	6.647 s	154.05 tok/s
7	8,192	1,024	739.41 ms	8.838 s	115.86 tok/s
8	8,192	1,024	735.99 ms	7.456 s	137.34 tok/s



Every request used the same 8,192-token input and 1,024-token output length. The best complete request reached **203.26 tok/s**, only 1.38% below 206.1. The slowest completed at 88.50 tok/s. The spread is primarily speculative-acceptance sensitivity rather than TTFT variation; TTFT stayed within 26.72 ms across all requests.

DSpark acceptance evidence - full-run accept length 2.5040

Observed accept length	Observed server decode throughput	Interpretation
5.10	265.67 tok/s	High acceptance; exceeds advertised reference
4.50	232.65 tok/s	High acceptance
3.52	183.59 tok/s	Moderate acceptance
3.17	165.51 tok/s	Moderate acceptance
2.52	131.49 tok/s	Near full-run TPOT-equivalent rate
1.50	about 78 tok/s	Low acceptance; speculative benefit collapses

3. Selected SGLang configuration

Component	Selected value
Target	RadixArk/Qwen3.8-27B-NVFP4
Target revision	554ebba9b5f1b79dc11246341960360e6ef05ef4
Draft	RadixArk/Qwen3.8-27B-DSpark
Draft revision	85ef153be924f17ce4bf62726954eeaa4a73e854
Speculative algorithm	DSPARK
Gamma / verify width	7 / 8 tokens
Target / draft attention	FlashInfer / FlashInfer
Hybrid linear attention	Triton decode, prefill, and verify; packed GDN decode enabled
Quantization	ModelOpt mixed NVFP4 W4A4 target; published DSpark draft
KV cache	FP8 E4M3
GDN state	FP32
Mamba strategy	extra_buffer
Mamba full-memory ratio	6.626953125
Chunked prefill	2,048 tokens
CUDA graphs	Target prefill, target verify, and draft verify captured
Reasoning / tools	qwen3 / qwen3_coder

Published runtime provenance

Artifact	Immutable identifier
Container	lmsysorg/sglang:qwen38-27b
AMD64 manifest	sha256:506525a5907ea22c9d445afb7c03603959b912de034d86915cf17da814f1a124
Source layer	sha256:923575b8eda6e97571097c2ee49cbdb96bfed39e7f86c3ec4fb24224f9be7764
dspark.py	sha256:def384fb88948dd7e29edbc941a1e4729fc134e1e82419858604ca2846be5c99
FlashInfer source	906181e3f4cf4bcc81835fb480db4011bbd80b62
FlashInfer cubin/JIT	0.6.18.dev20260807 / CUDA 13.0 cache

The exact published-image source was required because generic SGLang main initially failed at the quantized DSpark LM head. The image source contains the specialized SM120 and quantized LM-head path used by the successful candidate.

4. RTX PRO 6000 memory and capacity envelope

97,887 MiB	88,750 MiB	196,618	123
PHYSICAL VRAM	OBSERVED RESIDENCY	ALLOCATED KV TOKENS	MAMBA STATE SLOTS

Allocation

Allocation	Size	Notes
Target weights	21.68 GB	Qwen3.8-27B ModelOpt NVFP4
DSpark weights	2.71 GB	External draft model
Main SSM state	17.44 GB	FP32
Main convolution state	0.34 GB	FP32 state path
Intermediate SSM verification cache	28.12 GB	Eight-token DSpark verification
Intermediate convolution cache	0.23 GB	Verification window
Target KV cache	6.00 GB	3.00 GB K + 3.00 GB V
Draft KV cache	1.88 GB	0.94 GB K + 0.94 GB V
Available after pool setup	14.12 GB	Reported at initialization

Capacity boundary

Capacity field	Value
Checkpoint native context	262,144 tokens
Allocated KV-token pool	196,618 tokens
Maximum reported request input	196,612 tokens
Configured max running requests	48
Runtime Mamba-capped requests	24
State slots per request	5

The ratio 6.626953125 is calculated specifically for 8,192 input + 1,024 output tokens at concurrency 1 with FP32 GDN state and DSpark width 8. It prioritizes state capacity and leaves 196,618 KV tokens. Before advertising a fully live 262,144-token request, recalculate or lower the ratio and rerun long-context stability gates.

Power and thermals

Field	Value
Configured GPU power limit	500 W
Hardware default power limit	600 W
Clean-run peak power / temperature	Not captured; no unsupported estimate is reported

5. Functional gates and startup

Gate	Result	Elapsed	Evidence
Exact instruct	PASS	98 ms	QWEN38_TEXT_PASS
Thinking numeric	PASS	469 ms	37
Native screenshot vision	PASS	287 ms	Qwen3.8-27B identified
Structured tool call	PASS	308 ms	record_check(account=cash, amount=417)
Tool continuation	PASS	71 ms	AGENTIC_PASS
Systemd restarts	PASS	0	NRestarts=0

Cold-start timing

Stage	Time
Target + draft weight load	4.03 s
KV-cache allocation	0.164 s
Prefill graph capture	10.95 s
Target verify graph capture	3.40 s
Draft graph capture	0.92 s
Scheduler ready	22.21 s
Complete API readiness	26.27 s

Measurement integrity

Run	Output throughput	Disposition
Public port with unrelated long requests	102.81 tok/s	REJECTED
Loopback-only clean run	119.19 tok/s	ACCEPTED
Isolation improvement	+15.93%	No external fleet traffic

The rejected public-port run overlapped with unrelated 47K-56K-token requests and observed two running requests. The accepted run used loopback-only port 8010. The receipt's peak-concurrent field still reports 2 because the benchmark includes warmup/stream accounting; configured concurrency and effective measured concurrency were 1 and 0.9997, respectively.

6. Comparison and selected-default rationale

Reference	Rate	Difference vs clean average	Context
Clean DSpark full-run average	119.19 tok/s	baseline	8 random 8192/1024 requests
Social post	206.10 tok/s	+86.91 tok/s	Workload and acceptance unspecified
Best complete local request	203.26 tok/s	+84.07 tok/s	1,024 output tokens
Highest local decode window	265.67 tok/s	+146.48 tok/s	Accept length 5.10

The clean average is 86.912 tok/s, or 42.17%, below 206.1 and achieves 57.83% of that reference. This does not indicate a failed DSpark deployment: the same server exceeds 206 tok/s when acceptance is high. It shows that a single speculative-decode number is not portable across prompt distributions.

Different-engine community comparison

Engine / configuration	Baseline	Speculative	Gain
llama.cpp MTP depth 2	63.8	91.5 tok/s	+43.4%
llama.cpp MTP depth 4	63.8	85.7 tok/s	+34.3%
vLLM MTP depth 2	63.3	96.1 tok/s	+51.8%
vLLM MTP depth 4	63.3	116.3 tok/s	+83.7%

These qwen38-mtp figures use built-in MTP, other engines/checkpoints, and a different harness. They provide directional context only and are not an apples-to-apples DSpark comparison.

Selected default

Default decision field	Selected value
Public model ID	qwen3.8-27b
Serving engine	SGLang published Qwen3.8 image source
Target + draft	RadixArk NVFP4 + RadixArk DSpark
Service owner	qwen38-nvfp4-dspark-pro6000.service
Production endpoint	0.0.0.0:8000 after controlled promotion
Rollback	qwen38-abliterated-nvfp4-c6.service retained disabled
Required promotion gates	Cold discovery; text; reasoning; vision; tools; zero restarts; fleet check

7. Reproducibility and sources

Primary command shape

Benchmark argument	Value
Module	python -m sglang.bench_serving
Backend	sglang-oai
Dataset	random
Lengths	--random-input-len 8192 --random-output-len 1024 --random-range-ratio 1
Load	--num-prompts 8 --max-concurrency 1 --request-rate inf
Cache	--flush-cache

Source files

Artifact	Location
Raw benchmark receipt	D:\BlueStone\qwen38-release-kit\qwen38-nvfp4-dspark-pro6000-cookbook-isolated-8192x1024-c1.jsonl
Detailed Markdown report	D:\BlueStone\qwen38-release-kit\QWEN38-PRO6000-DSPARK-RESULTS-20260817.md
Candidate launcher	D:\BlueStone\qwen38-release-kit\serve-sglang-qwen38-nvfp4-dspark-pro6000.sh
Benchmark launcher	D:\BlueStone\qwen38-release-kit\benchmark-qwen38-dspark-published.sh

Online references

SGLang Qwen3.8-27B cookbook:
https://docs.sglang.io/cookbook/autoregressive/Qwen/Qwen3.8-27B

Live SGLang configuration source:
https://github.com/sgl-project/sglang/blob/main/docs/src/snippets/configs/Qwen/qwen3.8-27b.jsx

Community llama.cpp MTP comparison:
https://github.com/sudoingX/qwen38-mtp

Conclusion

The RTX PRO 6000 deployment is functionally sound, provenance-pinned, and measurably accelerated by DSpark when the draft is accepted. The defensible random-workload number is 119.19 output tok/s; 206+ tok/s is achievable in favorable decode windows but should not be represented as the universal average. The selected default preserves the exact tested parameters and retains the former abliterated NVFP4 service as a rollback.